

Younsoo Park

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Education

The Pennsylvania State University, University Park, PA Aug 2021 – May 2027 (Expected)
B.S. in Computer Science and minor in Mathematics Dean's List

Experience

SIGNUM (Nittany AI Alliance – PIT-UN Funded) — University Park Aug 2024 – Nov 2025
Co-founder of a \$3,750 PIT-UN-funded AI platform for hospital quality analysis and caregiver decision support.
- Built a multi-source ETL pipeline (CMS, NPPEs, Google Places) into a DuckDB warehouse and developed a Markov Transition Model for hospital star rating prediction with confidence intervals.
- Designed and deployed domain-specific RAG pipelines using FAISS and AWS Bedrock, reducing caregiver research time from 50+ hours/week to near-instant responses.

Republic of Korea Air Force — Korea, Republic of Sept 2022 – June 2024
- Promoted to Squad Leader; managed and coordinated 20+ airmen across high-tempo operational missions.
- Received a Commendation Award for sustained leadership performance and operational readiness.

Atom Tech Solutions LTD, Internship — Berkeley, CA May 2022 – Aug 2022
- Built secure login with SQL/JWT authentication, encrypted data storage, and automated credential generation.

Projects

Undergraduate Research Assistant & Team Lead Oct 2025 – Present
Reliable Federated TinyML Deployment for IoT Security (WIP, LCTES '26) — Sup. Dr. Suman Saha (Penn State) & Dr. Peilong Li (Elizabethtown College)

- Designed a federated learning pipeline (FedAvgM, 80 rounds, focal loss) on CIC-IDS2017; introduced server-coordinated cosine LR scheduling that improved attack recall from 46.7% to 93.85% and F1-score from 84.1% to 89.32%.
- Built a multi-stage compression pipeline (BatchNorm folding, knowledge distillation, structured pruning, INT8 QAT/PTQ) achieving $12.28\times$ model size reduction and 74.5% inference latency reduction for ESP32 deployment.
- Evaluated 48 compression configurations and conducted adversarial robustness analysis using PGD attacks (ϵ -sweep 0.01–0.2), identifying QAT trade-offs across moderate vs. aggressive compression regimes.

ASME @ Penn State Website Oct 2025 – Present
- Built the official ASME Penn State web platform using React 19, TypeScript, Vite, and Firebase; designed a 5-tier role-based permission system (Member to Admin) controlling content editing, approvals, and project management.

Hangukgwan (Family Business Web App) 2024 – Present
- Full-stack restaurant web app with 4-language i18n support (Korean, English, Simplified/Traditional Chinese), built with React + TypeScript, Node.js/Express, and MongoDB Atlas; integrated Google Maps API and JWT authentication.

Skills

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL

Frameworks & Tools: React, Node.js, Express, Flask, MongoDB, Firebase, Git, AWS Bedrock, DuckDB

AI/ML: TensorFlow, PyTorch, FAISS, SHAP, spaCy, Scikit-learn, Flower (FL)

Awards & Honors

- 1st Place, Project Competition – ThinkNeuro LLC April 8, 2026
- 1st Place, Battle Bots Competition – IEEE Student Chapter April 11, 2026
- PIT-UN Funding Award (\$3,750) – Nittany AI Challenge April 14, 2025
- 3rd Place, ACM Pathfinder Robotics Challenge April 12, 2025

Extracurricular

THON Technology Committee, Penn State: Technology Captain (2027) Apr 2026 – Present

Penn State Advanced Vehicle Team (AVT): Project Manager Fall 2026 – Present

Association of Computing Machinery(ACM): Web & Design Director Apr 2026 – Present

Lead Teaching Assistant, Math 230 (Multivariable Calculus) Aug 2025 – Present